Aadarsh Sahoo

Final Year Dual-Degree (B.Tech. + M.Tech.) Student
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Education

Indian Institute of Technology Kharagpur, India — Aug 2017 - Apr 2022
Major in Computer Science & Engineering, Dual Degree — GPA **9.36/10.00**

BJEM School, Odisha, India — Apr 2015 - Mar 2017
All India Senior School Certificate Examination, CBSE — Percentage Score **94.6%**

Interests

My research interests lie broadly in computer vision & artificial intelligence, specifically in learning algorithms using limited to no labeled data, leveraging self-supervision and multi-modalities in images and videos, with a goal to conduct fundamental research and design products benefiting humanity.

Publications

Aadarsh Sahoo, Rutav Shah, Rameswar Panda, Kate Saenko, Abir Das. *Contrast and Mix: Temporal Contrastive Video Domain Adaptation with Background Mixing.* Accepted at **NeurIPS**, 2021.
Aadarsh Sahoo, Rameswar Panda, Rogerio Feris, Kate Saenko, Abir Das. *Select, Label, and Mix: Learning Discriminative Invariant Feature Representations for Partial Domain Adaptation.* Under Review. Extended abstract accepted at the **NeurIPS DistShift Workshop**, 2021.
Aadarsh Sahoo, Ankit Singh, Rameswar Panda, Rogerio Feris, Abir Das. *Mitigating Dataset Imbalance via Joint Generation and Classification.* Accepted at the **ECCV Workshop (IPCV)**, 2020.

Internships and Research Experience

Boston University and UC Berkeley – DARPA LwLL — May 2021 - Aug 2021
Advisors: Prof. Kate Saenko (Boston University), Prof. Trevor Darrell (UC Berkeley)
- Worked on the video classification task for the DARPA LwLL Project, as part of the UC Berkeley team.
- Experimented with various algorithms to compete for the best performance in few-shot action recognition.
- Contributed a Spatio-Temporal Contrastive Learning algorithm leveraging temporal & spatial invariance in videos.
- Implemented codes in PyTorch and integrated it to a central framework developed and maintained by Kitware Inc.

CVIR IIT Kharagpur – Unsupervised Domain Adaptation in Videos — Jan 2021 - May 2021
Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)
- Worked on video domain adaptation for action recognition on various benchmark datasets.
- Proposed a framework using contrastive learning for domain alignment, background mixing for alleviating "shift".
- Leveraged video speed for domain invariance and target self-supervision for discriminability of the feature space.
- Implemented codes in PyTorch and performed experiments to achieve SOTA. Published the work at NeurIPS 2021.

CVIR IIT Kharagpur – Dynamic Source Selection for Partial Domain Adaptation — May 2020 - Oct 2020
Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)
- Proposed a dynamic source data selection, labeling, and mix-up based approach for partial domain adaptation.
- Leveraged domain discrepancy and Hausdorff distance in the feature space to learn the outlier distribution.
- Incorporated "selection" using a policy network for discrete binary decisions, optimized using Gumbel-Softmax.
- Implemented codes in PyTorch and performed experiments to achieve SOTA. Draft under review.

CVIR IIT Kharagpur – CycleGANs for Dataset Imbalance — May 2019 - Oct 2019
Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)
- Proposed an end-to-end CycleGAN-Classifier architecture to tackle dataset imbalance for image classification.
- Leveraged unsupervised image-to-image translation to design a dataset repairment strategy.
- Demonstrated improvement of Classifier & CycleGAN over prior methods using quantitative and qualitative metrics.
- Implemented codes in PyTorch and performed experiments to achieve SOTA. Published at IPCV, ECCV 2020.

Projects

Efficient Domain Adaptation through Knowledge Distillation Ongoing

Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)

- Working on efficient knowledge transfer algorithms across datasets possessing "domain-shift".
- Experimenting with a multi-subnet framework leveraging self-supervision techniques and knowledge distillation.
- Computation efficiency is achieved by performing budgeted and sample dependent computation at test-time.
- Implemented codes in PyTorch and and conducted various GPU-based experiments.

Universal Domain Adaptation through Adaptive Entropy-based Rewards Ongoing

Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)

- Working on a framework which uses a policy network to learn the distribution of unknown classes in Universal DA.
- Designed a sample-wise entropy-based reward along with a contrastive component to train the policy network.
- Used Gumbel-Softmax for discrete space optimization of the decisions taken by the policy network.
- Implemented codes in PyTorch and ran multiple GPU-based experiments.

Compiler for tinyC Course Project

- Designed and Implemented a compiler for TinyC (subset of the C language).
- Performed systematic and step-wise implementation of the phases involved like Lexer, Parser, Three Address Code Generator and Machine Dependent Code Generator.

KGP-RISC Microprocessor Course Project

- Designed a 32-bit Single Cycle Microprocessor for a given Instruction Set Architecture using Verilog and Implemented the Circuit in an Artix7 FPGA Board.

Bonafide: Plagiarism Detector Course Project

- Developed a Software using Java and Python Libraries which detects plagiarism in texts and documents.
- The development involved systematic and precise procedures according to SWE guidelines like preparing the SRS, implementing Use-Case diagrams, designing, UML diagrams, etc.

Activities

Teaching Assistant. CS60077 Reinforcement Learning, Autumn 2021, IIT Kharagpur.
Student Volunteer and Reviewer. CVPR-2021 workshop on Dynamic Neural Networks Meets Computer Vision.
Student Volunteer. Science Movement, Bhubaneswar.
Student Volunteer. National Service Scheme (NSS) IIT Kharagpur.

Honors & Awards

Secured a position in the Top-10; Granted Department Change, 2018. IIT Kharagpur, India.
All India Joint Entrance Examination, 2017. Joint Seat Allocation Authority (JoSAA), India.
Secured Rank 2 in Senior School, 2017. AISSCE, CBSE.
Regional Mathematics Olympiad (RMO) 2014. Homi Bhabha Centre for Science Education (HBCSE).
National Children's Science Congress State-level twice. Department of Science and Technology, Govt. of India.

Programming Skills

Languages: Python, C, C++, Matlab, LaTeX
Deep Learning: PyTorch, TensorFlow, Keras

Relevant Coursework

IIT Kharagpur: Probability and Statistics, Linear Algebra, Machine Learning, Deep Learning, Reinforcment Learning, Artificial Intelligence, Natural Language Processing, Scalable Data Mining, Algorithms, Formal Language and Automata Theory, Operating Systems, Computer Networks, Compilers, Computer Architecture, Discrete Structures.
Online MOOCs: CS231n: CNNs for Visual Recognition by Stanford, CS294-158: Deep Unsupervised Learning by UC Berkeley, Deep Learning Specialization by deeplearning.ai, Machine Learning by Stanford-Online.